UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

5054 PHYSICS

5054/02 Paper 2 (Theory), maximum raw mark 75

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1 **(a)** **(i)** weight of water (causes extra pressure)(**not** mass) B1

(ii) density of liquid/(sea-)water **or** gravitational field strength/acceleration of freefall (**not** gravity) B1

(b) **(i)** 3.6/3.60 × 10^5 Pa B1

(ii) $P_1 V_1 = P_2 V_2$ **or** $1.05 \times 10^5 \times 6000 = 3.60 \times 10^5 \times V_2$ C1
1700 **or** 1750 **or** 1800 cm^3 A1 [5]

2 **(a)** (WD =)Fx **or** 1680 × 50 C1
84 000 J A1

(b) friction/drag/resistance **of water/air** B1
work done against friction/resistance/drag **or** energy lost due to friction/resistance/drag **or** energy lost as heat/internal/thermal B1 [4]

3 **(a)** **(i)** (he) loses –ve charge C1
electrons lost (to surface) (positive electrons 0/2) A1

(ii) (becomes) negative/gains electrons B1

(b) **(i)** (he) discharges/(re)gains electrons/–ve charge (**not** current flow) B1

(ii) (Q =)It **or** 1.6 × 0.15 **or** 0.0016 × 0.00015 C1
2.4×10^n C1
2.4×10^{-7} C A1 [7]

4 **(a)** **(i)** one ray from M correctly reflected – checked by eye C1
two rays from M correctly reflected – checked by eye – **and** traced back to image A1

(ii) image point **clearly** marked at intersection/correct place checked by eye B1

(b) 0.34 m **cao** B1 [4]

5 **(a)** **(i)** C in correct position i.e. gap 4, 18 **or** 32 { **allow** arrows/ B1
R in correct position i.e. gap 11 **or** 25 { brackets < λ/2 B1
OR two correct positions but R and C reversed 1/2

(ii) 6.2 → 6.6 cm B1

(iii) (v =)fλ **or** 5.1/5100 × 6.4/0.064 (using candidate's **5 (a) (ii)**) C1
$3.16 – 3.37 \times 10^n$ C1
316 – 337 m/s A1

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(b) (longitudinal wave:) vibration/oscillation direction parallel to/in same direction as wave/energy travel direction (**not** L & R) B1
transverse wave: directions perpendicular **or** can be polarized (**not** up & down) B1 [8]

6 **(a)** **(i)** (I =)P/V **or** P = VI **or** 650/230 C1
2.8 **or** 2.83 A A1

(ii) 3, 4, 5, 6 **or** 7 A **only** B1

(b) **(i)** casing becomes live/at high voltage B1
current through user/user electrocuted/user shocked B1

(ii) fuse blows/melts/breaks B1
fuse in live wire/(microwave) disconnected/circuit broken/no current B1 [7]

7 **(a)** 1.(0) m B1

(b) **(i)** (for an object in) equilibrium/balance B1
$W_1x = W_2y$ (clear) **or** anticlockwise moment/torque/turning force = clockwise moment/torque/turning force B1

(ii) 18 000 × 1. 0 = T × 0.5 C1
36 000 N A1 [5]

8 **(a)** **(i)** 3 **cao** B1

(ii) 208 **cao** B1

(iii) 11 **cao** B1

(b) **(i)** 17 **cao** B1

(ii) 20 **cao** B1 [5]

[Total: 45]

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Section B

9 (a) (i) 100 – 22 **or** 78 — C1
(Q =)mcΔT **or** 35 × 4200 × 78 — C1
1.1/1.1466/1.15 × 10^7 J — A1

(ii) (t =)E/P **or** P = E/t **or** 1.15 × 10^7/2600 — C1
4.4/4.41/4.42 × 10^3 s — A1

(iii) heat escapes/lost (to kitchen) **or** heat to heat the boiler/heater **or** not all heat ends up in water **or** heat to cause evaporation **or** used as latent heat (**not** heat wasted) — B1 [6]

(b) (i) hot/warm **water** expands (**not** molecules expand) — B1
density (of hot/warm water) decreases — B1
hot/warm water rises — B1
convection current/circulation **or** cold water sinks — B1
mixes water (max 4) — B1

(ii) metal/steel is (good) conductor/poor insulator **or** plastic is poor conductor/ insulator — B1
more heat transferred through steel/less through plastic **or** heat transferred more quickly through steel/less quickly through plastic — B1 [6]

(c) (i) evaporation **OR** condensation — B1

(ii) any **two** points

only occurs at surface		boiling needs heat/	
occurs at any temperature		condensation releases heat	B1
produces cooling		boiling: liquid to gas/	
no bubbles	B2	condensation: gas to liquid	B1 [3]

[Total: 15]

10 (a) (i) (W =)mg **or** 0.5 × 3.7 — C1
1.8/1.85/1.9 N — A1

(ii) 3.7 m/s^2 **not** N/kg — B1

(iii) (KE =) ½mv^2 — C1
½ × 0.50 × 3.2^2 — C1
2.6 **or** 2.56 J — A1 [6]

(b) (i) A compares/measures (unknown and known) masses/amount of matter — B1
B measures/is dependent on weight/force of gravity (and hence mass obtained) — B1
Mars weights/forces of gravity are less than/different from (Earth) — B1

(ii) A **or** lever arm balance **or** balance with discs — B1 [4]

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(c) (i) volume B1

(ii)

Either:	**Or:**
record value of water in measuring cylinder (**not** beaker)	water/liquid in displacement/eureka can
insert rock	full to overflowing
record new value	immerse rock
subtract (to obtain volume) **or** measure rise)	(new) overflow in measuring cylinder (**not** beaker is volume)
m/volume **or** *m*/subtraction (max 4)	*m*/volume **or** *m*/subtraction (max 4)

B4 [5]

[Total: 15]

11 (a) (i) (I =)V/R **or** V = IR (in **(i)**/**(ii)**) **or** 9.0/20 (in **(i)**) **or** 0.45 × 16 (in **(ii)**) C1
0.45 A A1

(ii) 7.2 V (max 3 for **(i)** and **(ii)** together) A1 [3]
C1 may be awarded for **either** A mark

(b) (i) R → T **and** line of positive slope throughout B1
straight line, positive intercept on R-axis **and** slope/0 on **kelvin** scale B1

(ii) voltmeter reading falls B1
current (supplied by battery) falls **or** X takes greater proportion of p.d. **or** 16 Ω takes smaller proportion of p.d. B1

(iii) 0 **and** to/→/- B1
8/9/10/whole number not greater than 20 V (usual unit penalty) B1 [6]

EITHER:

(c) (i) use small, metal conductor as probe/sensor **or** calibrate V reading (with known T) B1
the voltmeter reading is used to find T B1

(ii) any **two** from: high temperatures /remote reading/robust/quick acting/direct input to computer/low heat capacity B2

(iii) equal changes in one/T do **not** produce a equal changes in the other/V **or** graph with axes labelled **not** straight **or not** proportional to B2 [6]
not a straight line **or** not same change **or** change in V is not uniform 1/2

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OR:

(c) (i)

In		Out		
0	0	1	} all correct	B1
0	1	1		
1	0	1		
1	1	0	correct	B1

(ii) both inputs = 0 B1
output = 1 B1

(iii) A **and** B inputs = 1 B1
output = 0 B1 [6]

[Total: 15]

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